

Evaluation of the Compressive Strength of Concrete

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Background

Concrete plays a key role in modern construction, particularly in load-bearing structures such as foundations, columns, beams, and slabs. Its extraordinary compressive strength is essential for withstanding the demands of construction. Compressive strength, defined as the applied force at failure divided by the cross-sectional area, makes concrete a go-to material for ensuring structural integrity.

The composition of concrete typically consists of cement, water, and a blend of sand, gravel, or crushed stone. The meticulous combination of these materials is crucial for achieving the desired compressive strength. Additionally, the age of concrete is a significant factor in predicting its strength. Predicting concrete strength accurately is essential for construction planning. Having reliable predictions based on the mixture and age of the concrete is invaluable for ensuring that structures are subjected to loads only when the concrete has reached optimal strength. This ensures the safety and longevity of the constructed infrastructure.

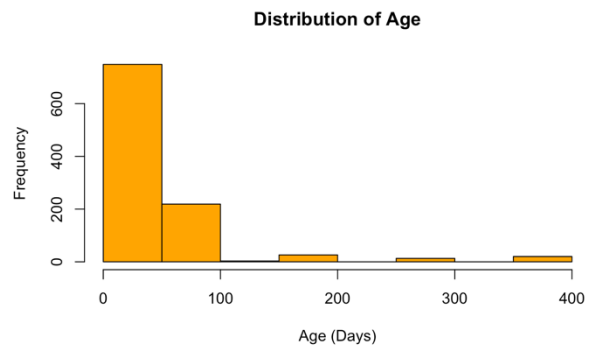
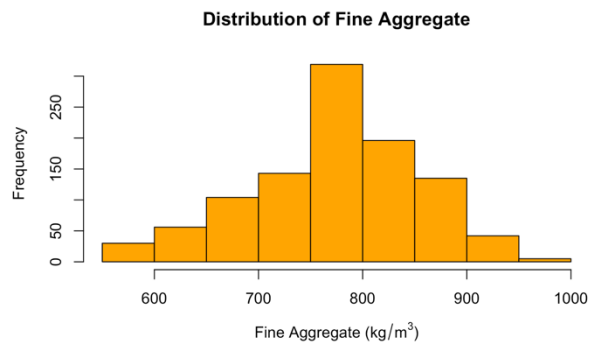
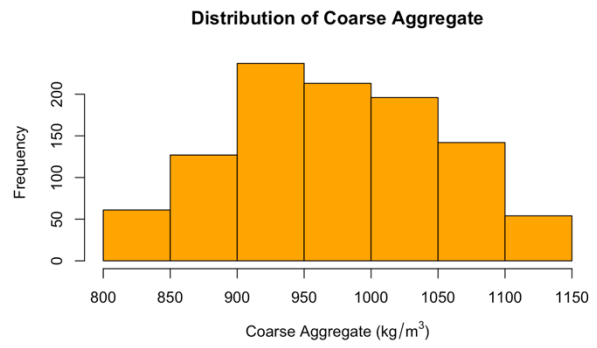
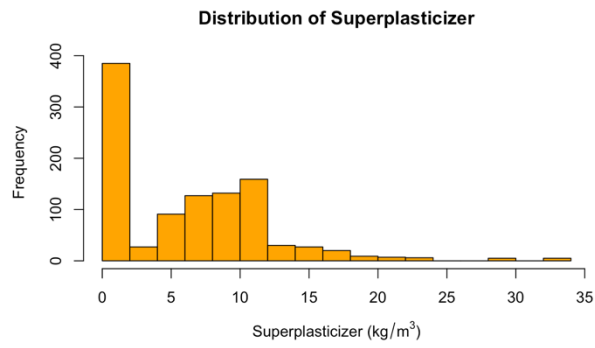
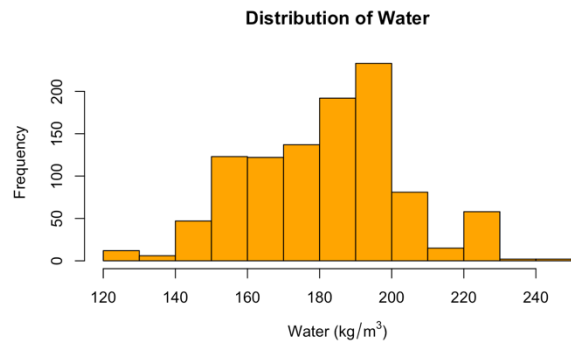
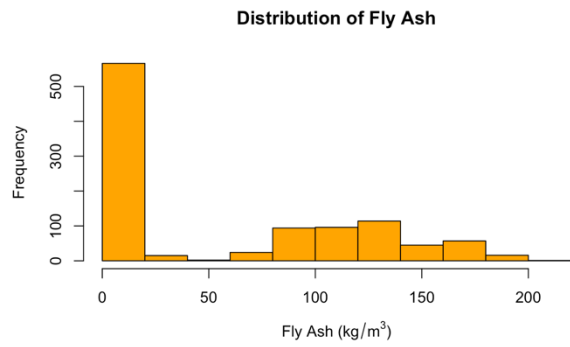
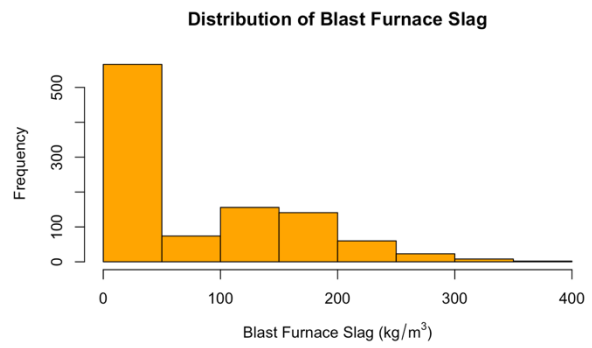
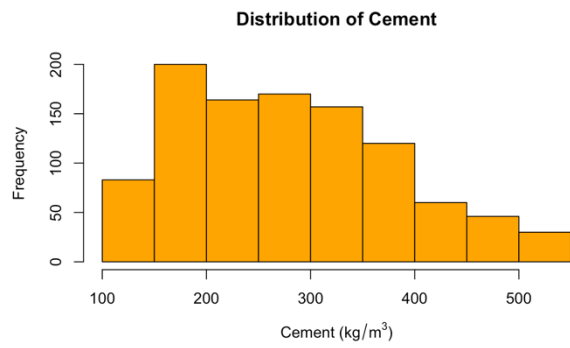
The goal of this study is to examine the relationship between the concrete mixture components and compressive strength. By exploring both linear and nonlinear regression models, we aim to optimize concrete mixtures for maximum strength while balancing interpretability and accuracy.

Data

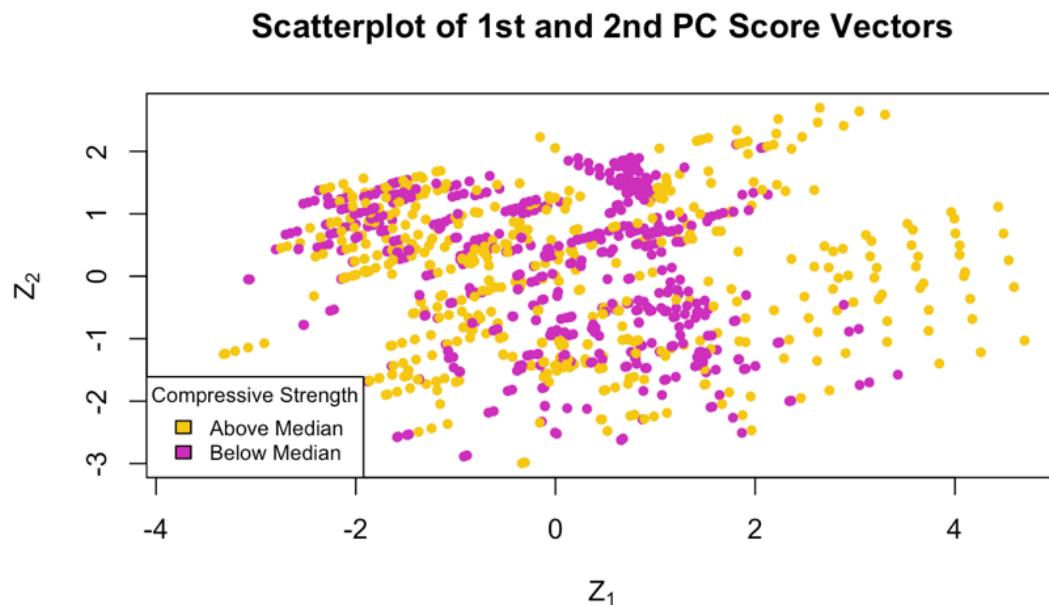
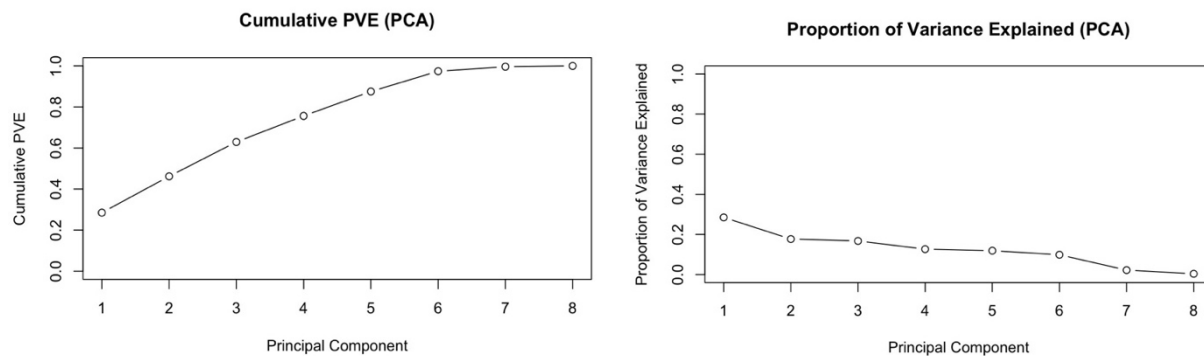
The data for this project is from the UC Irvine Machine Learning Repository. Each observation is a concrete mixture produced in a laboratory. The mixtures consist of up to 7 ingredients, each of which is recorded in kilograms per cubic meter (kg / m^3). The mixtures also have an age component, measured in days. Therefore, our data has 8 numeric predictor variables. The target variable is the compressive strength of the resulting concrete, which is measured in megapascals (mPA.)

Exploratory Data Analysis

We began by performing exploratory data analysis. First, we plotted a histogram of each predictor variable. These plots are given on the next page. The cement, water, coarse aggregate, and fine aggregate variables are distributed symmetrically. In contrast, the blast furnace slag, fly ash, superplasticizer, and age variables are heavily right skewed. They are right skewed because some of the observations have values equal to zero. This tells us about how the instances were produced in the laboratory. For some trials, the experimenters excluded certain ingredients. When an ingredient was excluded, they recorded a value of zero for that variable. Because of this, there is not much we can do to remove the skew. Even if we transform the distribution, there will still be observations stacked at the same value on the x-axis. The only variable that could successfully be transformed is the age variable. This is because the values are spread over a wider range. That is, the values are not stacked at zero like in the other examples.



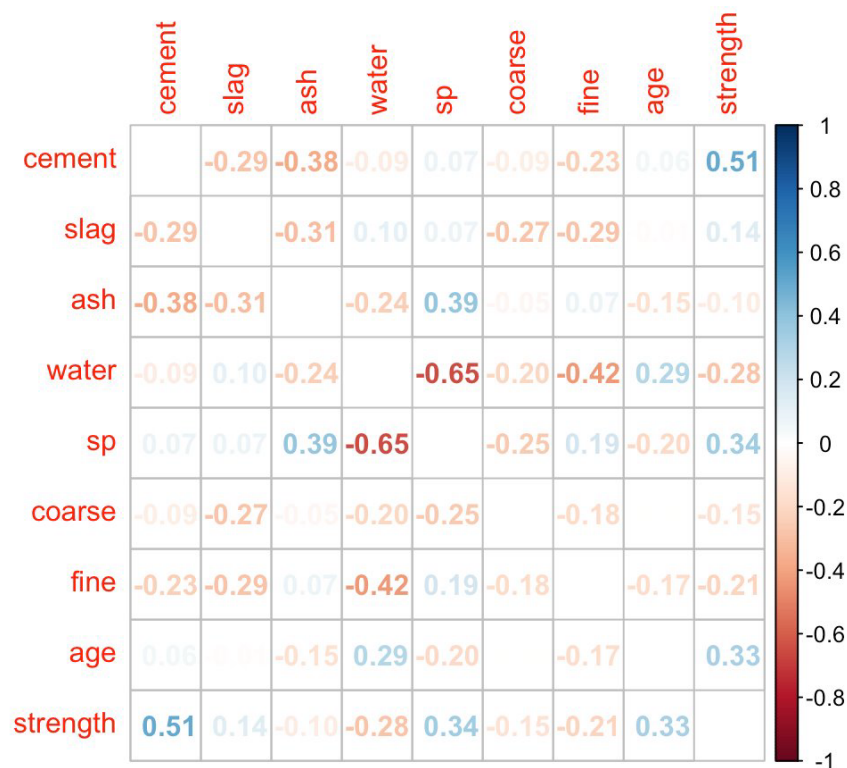
Next, we performed principal components analysis on the 8 predictor variables. Our goal was to reduce the space to two dimensions so that we could visualize all the predictors at once and potentially find some enlightening pattern. The results are shown in the three plots below. The top plot gives the proportion of variance explained by each component. The first principal component only explains about 30% of the variance in the data. The middle plot gives the cumulative proportion of variance explained by the principal components. The first 6 principal components combine to explain almost all the variation in the data. The last two principal components do not contribute much. Our main interest lies in the third plot at the bottom of the page. Here, we plot the first two principal component score vectors. We colored the points based on whether the corresponding concrete observation has a strength above or below the median strength. Unfortunately, there are not any interesting patterns in this two-dimensional visualization of the predictors.



Multiple Linear Regression

In this section, we begin the supervised learning portion of the project. We start by predicting concrete strength using multiple linear regression. Although we did not include them in the report, we created scatterplots of each predictor against the target variable strength. These plots are in the R file. There is a clear non-linear relationship between cement and strength. There is also a non-linear relationship between age and strength. Linear regression assumes a linear dependence between the predictors and target variable. Therefore, not all the assumptions for linear regression are met. For the sake of completeness, we continued with the process anyway. However, we exercised caution when drawing conclusions since the results might not be accurate with the broken assumptions.

In the plot below, we check for correlation between the predictors and target variable. The cement variable has a moderate positive correlation with the target variable strength. The sp and age variables also have positive correlations to strength, but these correlations are a bit weaker. As for collinearity, the water and sp variables have a moderate negative correlation. This should not be too much of an issue, as the correlation is not extremely strong.



After that, we checked for multicollinearity among the predictors using the variance inflation factor (VIF). Some of the variables have a VIF above 5, but none of them have a VIF above 10. Thus, multicollinearity is not an issue. The age variable has the lowest VIF, and the ash variable has the highest.

	Cement	Slag	Ash	Water	SP	Coarse	Fine	Age
VIF	7.09	6.96	7.75	6.77	2.94	5.06	6.70	1.12

Ultimately, we chose to test 4 different linear regression models. First, we fit a full model that included all 8 predictors. Then we tested a smaller model obtained using best subset selection. Finally, we tested the effectiveness of ridge and lasso regression.

Linear Models			
Type	Training MSE	CV MSE	Test MSE
Full Model	108.12	112.01	106.77
Best Subset Model	108.7	111.64	109.07
Ridge Regression ($\lambda = 0.85$)	110.27	113.97	108.85
Lasso Regression ($\lambda = 0.1$)	108.36	112.76	107.3

The adjusted R-squared value for the Full model is 0.6179. Based on the p-values, cement, slag, ash, water, and age are identified as the most significant predictors. Although the Test Mean Squared Error (MSE) was lower than the Training MSE, this trend is not observed during 10-fold cross-validation. The strangely low Test MSE is likely a result of the random split between the Training and Test Set. The cross-validation MSE is a better estimate of generalization performance.

When assessing the Best Subset model, the adjusted R-squared, Cp, and BIC statistics were utilized. The results show a model with six variables is suggested by the adjusted R-squared and Cp statistics. However, the BIC statistic favors a smaller model comprising five variables. Notably, the discrepancy in adjusted R-squared between the five- and six-variable models is minimal. The same trend is observed with the Cp statistics. Hence, we opted for the simpler five-variable model, as recommended by the BIC plot. These variables include cement, slag, ash, water, and age, all of which also exhibited the most significant p-values in the full model. The Best Subset model has a training error which is higher than for the Full model, but when running 10-fold cross validation this reverses with the Best Subset model having a lower MSE than the Full model.

For the Ridge Regression Model, cross-validation tells us that the best value of lambda is $\lambda = 0.85$. This is a low weight for the penalty term meaning that the coefficient values are not that different from the coefficients in the full model. The Lasso Model also has a low λ , with $\lambda = 0.1$ as the best weight for the penalty term. The Lasso Model is slightly better than that of the Ridge Model with a CV MSE of 112.76 as compared to 113.97, but both are still worse than the Full Model. This suggests that the decrease in variance provided by regularization did not offset the corresponding increase in bias.

Non-Linear Models			
Type	Training MSE	CV MSE	Test MSE
Polynomial Regression	33.96	38.33	42.07
Cubic Spline	24.78	33.37	32.6
Natural Spline	25.66	32.21	34.2
Local Regression	20.85	30.03	29.63
Polynomial Kernel SVM	59.24	89.71	67.88
Radial Kernel SVM	17.47	33.65	32.41
Random Forest	6.02	33.85	29.67

Non-Linear Regression

So far, many of our tests have indicated that there is non-linearity in the relationship between the predictors and the target variable. In this section, we explore various non-linear regression models to predict the concrete strength from the mixture ingredients. These models are generalized additive models that include all 8 predictors. The additive assumption is that one variable's effect is not influenced by the other's levels. Since we observed little collinearity and multicollinearity between the predictors, we believe this assumption holds. The non-linearity in our generalized additive models comes from the basis function applied to each predictor. Note that the basis function is the same for each variable. For example, we start with polynomial regression. The polynomial basis function is applied to each predictor in the model, and each basis function uses the same degree. This concept is made clear in our R file. After polynomial regression, we tested cubic spline basis functions, natural spline basis functions, and local regression basis functions. We concluded by trying some regression extensions for typical classification models. Specifically, we tested the effectiveness of support vector machine and random forest regression.

Polynomial Regression model				
Degree	2 nd	3 rd	4 th	5 th
Training MSE	63.38	41.66	33.75	32.06
Validation MSE	67.47	54.96	38.33	39.92

Among four different degree options, the Polynomial Regression model with a degree of 4 exhibits the lowest MSE, indicating it as the preferred choice for this dataset according to 10 - fold cross-validation. Here we see a trend that as degrees increase the training MSE decreases. But when looking at the 10-fold cross-validation MSE we notice that the model starts to overfit and because of that the validation MSE starts to go back up after the fourth degree.

Cubic Spline model								
df	5	6	7	8	9	10	11	12
Train MSE	31.2097	29.2460	26.3994	25.8772	25.9097	24.4254	23.9601	23.4452
Valid. MSE	40.0017	37.3842	34.7467	33.9889	35.0856	33.3725	33.8342	33.6221

For the Cubic Spline model, 10 degrees of freedom (df) is the optimal choice based on the 10-fold cross-validation MSE values. Here we can see that as the degrees of freedom increase the Training MSE decreases. For the 10-fold cross-validation MSE, that does not appear to be the case. It shows an overall trend of decreasing, but the 9th degree of freedom is an outlier where it increases significantly before going down again and leveling off with small increases.

Natural Spline model								
df	4	5	6	7	8	9	10	11
Train MSE	35.6251	33.6653	32.6208	30.7630	26.6933	25.4058	24.6459	24.2206
Valid. MSE	39.6130	39.1753	38.9958	38.3283	33.3145	32.2141	32.6980	32.5437

Among these options for the Natural Spline model, the model with 9 degrees of freedom demonstrated the best generalization to new data, yielding a 10-fold cross-validation MSE of 32.21. Here there is a general trend in both Train and Validation MSE to decrease as the degrees of freedom increase. However, after 9 degrees of freedom, the cross-validation MSE begins to rise due to overfitting. The performance for the Natural Spline model is marginally better than with the Cubic Spline model.

Local Regression model								
Span	.15	.20	.25	.30	.35	.40	.45	.50
Train MSE	20.3536	22.5542	24.6606	27.1013	29.4305	31.5052	34.3766	37.7105
Valid. MSE	30.0279	30.6500	31.5140	33.1758	35.5400	36.8450	39.3018	42.4275

Here, we use Local Regression basis functions and conduct 10-fold cross-validation to determine the optimal span. Among these spans, a span of 0.15 yields the best validation MSE. Attempting to use even smaller spans resulted in errors. For Local Regression, we see that as we increase the span, both Train and Validation MSE increase. Showing that the lower Span is the way to go for Local Regression Model. This cross-validation error is better than the Cubic and Natural Splines models.

Polynomial Kernel SVM										
Cost	25	10	5	1	.5	.1	1	.5	1	.5
Degree	3	3	3	3	3	3	5	5	4	4
CV MSE	89.71	89.72	91.32	93.23	97.96	129.50	140.86	144.99	148.47	158.09

Here, we move on to support vector machines. We used cross-validation to determine the optimal cost and degree hyperparameters for the Polynomial Kernel SVM. We can see that as the cost increases the cross validation MSE increases and as degree increases the cross validation MSE increases. Thus, making the best cross-validation error 89.71, with degree 3 and cost equal to 25. However, it is worth noting that compared to some of the other non-linear models tested, this MSE is high.

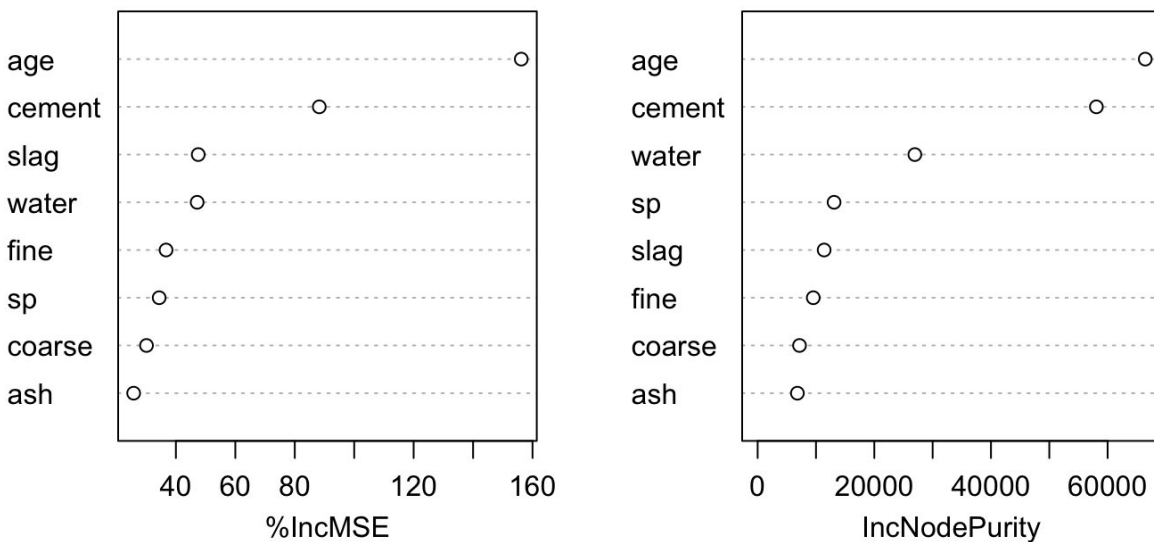
Radial Kernel SVM										
Cost	10	25	50	100	25	50	10	100	5	10
Gamma	.05	.05	.05	.05	.1	.1	.1	.1	.1	.5
CV MSE	40.17	36.43	33.64	34.15	34.30	34.52	34.28	36.62	37.78	41.32

For Radial Kernel SVMs we notice the trend that occurred in Polynomial Kernels for cost does not occur here regarding the effect with MSE. Initially, as cost goes up, we see that cross validation MSE decreases but then MSE starts to increase as cost goes up once it hits a point. For gamma, we did not notice any trend regarding the effect on MSE. According to the cross-validation results for Radial kernel SVM, a cost of 50 and a gamma of 0.05 generalize the best. The CV MSE of 33.64 is similar to the errors we received from the Splines models.

Random Forests Best Number of Variables to Consider at Each Split								
P	1	2	3	4	5	6	7	8
Training MSE	10.11456	10.01863	10.01491	10.15732	10.03588	10.11296	10.01385	10.08161
Validation MSE	33.94489	33.66857	33.86032	34.02237	33.85326	33.96069	34.03316	33.93944

Finally, we test the effectiveness of random forest regression. For our Random Forest Model, selecting from a sample of 5 variables at each split yields the most favorable results, with a corresponding 10-fold cross-validation MSE of 33.85. This value is close to the MSEs obtained from Spline models and Radial SVM. Among different numbers of variables considered at each split (p), the validation MSEs range from 33.66857 to 34.03316, with p=5 demonstrating the lowest error.

Variable Importance For Concrete Mixture



Here the Random Forest model is used to determine variable importance. The left plot illustrates the increase in error when each variable is excluded from the model. Clearly, age is the most important variable, as excluding it from the model results in a 160% increase in MSE. The cement variable also appears to be important. The right plot shows the average increase in node purity when each variable is used at a split in a tree. Since we are dealing with regression trees, node purity is measured by training residual sum of squares. Like the previous graph, this plot tells us that the age and cement variables are the two most important variables.

Conclusion

Based on the analysis of the dataset, several key observations can be made. Firstly, it is important to note that multicollinearity and correlation did not pose any significant issues, indicating that the variables are independent. However, the linear model broke the assumption of linear dependence between the predictors and response. Consequently, the best-performing linear model was a 5-variable model, which achieved a terrible CV MSE of 111.64. Ridge and Lasso regression techniques did not improve upon this result since regularization cannot help models that underfit the data.

Contrary to the linear models, non-linear models demonstrated superior performance, indicating that the relationship between predictors and the target variable is complex and non-linear. Among these models, Local Regression emerged as the top performer, with a CV MSE of 30.03, highlighting the effectiveness of capturing non-linear relationships.

It is also important to observe that age and cement played an influential role in determining the final concrete strength. In conclusion, the analysis emphasizes the importance of considering non-linear relationships and selecting appropriate modeling techniques to achieve the best predictive performance in this dataset.

One thing not in the variables that could have had an impact on the relationships between the components is environmental factors such as temperature and moisture, which can impact the properties of concrete. Changes in things like these environmental factors can also contribute to nonlinear effects seen in this study on concrete strength.

References

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